Minimised Evidence Sharing for Cross-Bank Money-Laundering Detection: Pre-Registered Sealed Tests and Fresh Replication on AMLworld

Allaudin Ansari
Affiliation
City, Country
email

Ravi
Affiliation
City, Country
email

***Abstract*—Money-laundering schemes are spread across banks, so each bank's monitoring sees only a fragment and rarely links the parts. We study a protocol in which banks exchange *minimised evidence* about counterparties—a calibrated score bucket and bucketed behaviour counters, keyed by pseudonymous tokens from an oblivious pseudorandom function—and evaluate it under a discipline uncommon in this area: every hypothesis is pre-registered, decided once on a sealed test, and then confirmed on two further test windows that were never opened during development. On AMLworld, sharing raises linked scheme detection from 0.063 to 0.534 on the high-illicit fresh test (gain 0.47, 95% CI [0.43, 0.51]) and recovers 0.92 of the gain of a pooled relational oracle; the gain holds on all four sealed tests, without suspected dataset artefacts, and on a base detector that outperforms published gradient-boosting results on the standard benchmark. Analysts review 6.8–8.5× fewer alerts per linked scheme. A leakage audit with explicit attacks finds that a bare alert flag reveals high fan-in in the low-illicit regime; a gated flag, chosen on validation by a frozen rule, removes the leak on fresh data at no detection cost. We report the claims that failed—typology attribution and the token re-identification target—alongside those that held.**

***Index Terms*—anti-money laundering, inter-bank information sharing, privacy-enhancing technologies, oblivious pseudorandom functions, pre-registration, leakage audit**

I. Introduction

Laundering schemes deliberately cross institutional boundaries: funds are gathered at one bank, layered through accounts at others and paid out elsewhere. Each bank's transaction monitoring sees only the legs that touch its own customers. The result is a familiar pattern in practice—many isolated alerts, few connected cases—and a familiar question: how much would banks gain by sharing what they know, and what would sharing reveal?

Prior work answers the first half with federated learning, secure computation or supervisory analytics (Section II), usually on one data split chosen and reported by the same team, and answers the second half by the properties of a mechanism rather than by measuring what a curious peer can infer. We take a different route on both.

Evidence, not models. When a bank's detector flags an account, it can ask the banks that hold the account's counterparties for a minimised evidence record about them: a score bucket and bucketed behaviour counters, keyed by a token that only an online, metered, logged service can produce. A ladder of sharing levels (S0–S4, plus progressive disclosure) separates what each extra field buys from what it reveals.

Evaluation as a sealed test. Hypotheses, metrics, operating points and model settings were registered at the outset; provisional thresholds were fixed once, from validation, at a pre-planned gate, and everything was frozen before the test was opened [1]. Each frozen configuration was evaluated once on a sealed test, with every opening logged. Improvements made after that test were developed on validation data only, frozen by written selection rules, and confirmed on two further 16-day windows of AMLworld [2] that had never been opened.

Contributions.

- A minimised evidence-sharing protocol for cross-bank detection, with OPRF tokens [3], a fusion layer and a pooled linking rule, and a ladder of sharing levels that isolates the value of each disclosure.
- A pre-registered evaluation on four sealed tests, two of them fresh: sharing raises linked scheme detection by 0.38–0.57 over local monitoring and recovers 0.64–0.98 of a pooled relational oracle's gain; the gain survives removing suspected artefact features and replacing the detector with one that outperforms published gradient-boosting baselines.
- An empirical leakage audit that found, and a pre-registered fix that closed, an attribute leak through alert flags.
- Case-quality measures that matter to investigators: more complete cases and 6.8–8.5× fewer alerts per linked scheme.
- A negative result on narrative generation with small local language models, and a select-then-template design that is grounded by construction.

II. Related Work

Collaborative detection. Federated learning lets banks exchange model updates instead of data: Suzumura et al. [4] proposed federated graph learning across banks, and Zheng et al. [5] train a federated graph network on country-level subgraphs of AMLworld. The US/UK PETs Prize financial-crime track produced systems that bring a bank's private account flags into a payment hub's model: with MPC and federated learning [6], private set intersection and secure aggregation [7], or differential privacy with secure summation [8]. Secure computation has been used to propagate risk scores across banks [9] and to trace money flows at national scale under homomorphic encryption [10]; Project Aurora [11] compared siloed, national and cross-border collaborative analysis on synthetic payments.

What differs. Table I compares these works. We share evidence about counterparties rather than models, and we measure leakage with explicit attacks instead of relying on a mechanism's guarantee alone. None of the works above reports results on a pre-registered, sealed test, and several select models on the split they report; none replicates on data never opened during development.

Benchmarks. AMLworld [2] provides synthetic transactions with laundering labels and typologies. Multigraph GNNs [12] and subgraph features with gradient boosting [13] set the transaction-level state of the art; we anchor our detector against those published numbers.

III. System

A. Setting and threat model

A federation of banks each sees the transactions with one end at its own customers, including the counterparty's account and bank, and nothing more. Adversaries are honest-but-curious: a peer bank follows the protocol but inspects every token, request and response it receives, and holds candidate account lists from its own payments; the token service sees blinded queries and does not collude with banks. Poisoning, collusion and compromised hosts are out of scope. No legal or regulatory claim is made.

B. Local detection (S0)

Each bank scores every active account in every 6-hour window with a LightGBM model [14] on 47 features in four families (behaviour, graph, history, neighbourhood), each computed from the current window and the previous 72 hours of the bank's own view. Scores are calibrated per bank, and each bank alerts a fixed number of account-windows per window (Section IV), so sharing cannot raise recall by raising alert volume. A bank links its own alerts into incidents when the accounts transacted directly within 72 hours.

C. Tokens

Banks must recognise that the account a peer asks about is one they paid, without exchanging account numbers and without holding a key that tokenises arbitrary identifiers. Each bank blinds an identifier and sends it to a token service, which evaluates a pseudorandom function under a per-epoch key (one epoch is seven days) without seeing the identifier, following the OPRF construction of RFC 9497 [3]. Tokenising a candidate therefore needs an online, logged query, and each bank has a per-epoch budget, sized to twice the largest legitimate use on validation. Shared-key hashing is kept only as a positive control in the audit.

D. Evidence and sharing levels

An account-window above the evidence floor (calibrated probability ≥ 0.01) or alerted asks, for each counterparty at another bank within the 72-hour lookback, what that bank holds. The responding bank answers according to the sharing level:

- **S1**: whether it has alerted the counterparty (alert flag), with a time bucket.
- **S2**: a minimised record—probability bucket; up to three deviating features with direction and a coarse magnitude; bucketed fan-in, fan-out and pass-through, and cross-bank, new-counterparty and amount counters. No raw values.
- **S3**: S2 with every field except the time bucket and flag released through k-ary randomised response [15, 16] at a total $\varepsilon \in \{0.5, \ldots, 8\}$ per record, capped per account and epoch.
- **S4**: S2 plus an 8-dimensional autoencoder embedding (a leakage upper reference).
- **PD**: progressive disclosure—the S1 answer, plus the S2 record only when the requester's own score bucket is at least $\tau \in \{2, 3, 4\}$.
- **Gated flag** (S1F, added in Section IV-D): the S1 flag is returned only when the requester's score bucket is at least 2.

E. Fusion and linking

Each bank combines its own score with the returned evidence (counts of requests, flags and records; the highest and summed probability buckets; fan and pass-through buckets) in a per-bank logistic stacker, cross-fitted on the two halves of validation. Banks then cut their own queues at the same alert budget, and alerts are linked across banks when their accounts transacted within 72 hours (the pooled linking rule). Two pooled references bound the ladder: the *oracle*, one LightGBM on every bank's data, and the *relational oracle*, which adds to each pooled score its counterparties' pooled scores across the whole federation. The relational oracle is the ceiling in all ratios below.

F. Case narratives

A gradient-boosted attributor assigns a typology to each linked case from the fused evidence (or leaves it unattributed below a confidence threshold). A local language model then writes a short narrative under a JSON schema whose typology must equal the attributed one and whose cited fields must exist in the evidence; failed checks trigger one retry, then a fixed template. The select-then-template variant (N4) asks the model

TABLE I
COLLABORATIVE ANTI-MONEY-LAUNDERING AND FINANCIAL-CRIME DETECTION: WHAT IS SHARED, HOW IT IS PROTECTED, AND HOW IT IS EVALUATED

Work	Shared between institutions	Protection	Data	Evaluation	Privacy evidence
Suzumura et al. [4]	Model parameters (global graph features planned)	Federated learning; DP/MPC proposed, not evaluated	Simulated, six banks (FCA TechSprint)	One test split, prototype	None
Arora et al. [6]	Bank account flags, inside a payment hub's training	Federated learning + MPC (oblivious transfer, secret sharing), noise	PETs Prize synthetic payments	Challenge test environment	Design argument
Abadi et al. [7]	Identity alignment and model updates	Private set intersection + secure federated learning	Synthetic payments	Scalability and accuracy, prototype	Security proof
Khan et al. [8]	Embeddings and gradients	Differential privacy + secure summation	PETs Prize synthetic, AMLSim	One test split; best test epoch reported	Formal DP bound
van Egmond et al. [9]	Encrypted risk scores, propagated across banks	Secure multi-party computation	One bank's transactions	Precision at fixed recall	Design argument
Brand et al. [10]	Encrypted flow-tracing indicators	Homomorphic encryption	Nation-scale deployment	Run time only	Design argument
BIS Innovation Hub [11]	Collaborative analysis and learning	Privacy-enhancing technologies	Synthetic, national and cross-border	Scenario comparison	Not reported
Zheng et al. [5]	Model updates and aggregated node embeddings	Federated learning (no formal guarantee)	AMLworld HI-Small subset [2]	One split (5% train)	None
This work	Minimised evidence records keyed by OPRF tokens; alert flags, gated	Pseudonymous tokens (OPRF), data minimisation, flag gating; LDP and progressive disclosure as options	AMLworld HI/LI Medium and two never-opened Large windows [2]	Pre-registered sealed tests, fresh replication, 95% CIs, Holm	Empirical attacks (4 targets) and token re-identification

TABLE II
TEST SETS

Test set	Transactions (millions)	Laundering rows	Schemes	Cross-member test schemes
HI-Medium	31.9	31,146	2,756	480
LI-Medium	31.3	15,270	456	91
HI-Large	30.3	39,798	6,638	559
LI-Large	29.7	18,037	896	78

Each set is a 16-day dense window of AMLworld, split by 6-h window 60/20/20 into train, validation and test with one-window gaps; eight institutions (federation B8) hold every account. Cross-member test schemes: laundering schemes in the test partition that touch two or more institutions. Medium: the week-18 sealed tests; Large: the A25 fresh tests.

only to choose up to four evidence fields; fixed templates write every sentence.

IV. EVALUATION METHOD

A. Data and federation

We use the IBM AMLworld synthetic data [2] in its high-illicit (HI) and low-illicit (LI) variants. Table II lists the four sealed tests. For each set, a label-free rule finds the dense period of transactions and keeps 16 days (Medium: the whole dense period; Large: the last 16 days of its dense period). Windows are 6 hours; train, validation and test take 60/20/20% of the windows, with a one-window gap at each boundary. All banks are grouped into eight institutions balanced by accounts (federation B8), so every account is held by a member.

B. Unit, labels and metrics

The unit is the account-window: an account held at a member in one window with at least one transaction. It is positive if it touches a laundering leg of a scheme assigned to its own partition; legs of schemes assigned to another partition, and busy accounts that a scheme crossed once (at least 20 legs, under 10% laundering), are masked. Each bank alerts one account per 1,000 accounts per day, exactly. A scheme is *detected* if any of its account-windows is alerted within 72 hours of its first window, and *linked*—the primary metric—if at least two of its accounts fall in one incident within that time. Before any sharing level was built, a gate required the oracle to beat S0 by at least 0.05 in linked detection on validation: it did by 0.508 [0.465, 0.550] on HI-Medium, almost entirely through linking (any-alert detection: S0 0.864, oracle 0.886).

C. Pre-registration and the sealed test

Hypotheses (Table III), operating points, model settings and statistical rules were registered at the outset. Thresholds marked provisional could be revised once, from validation, at a pre-planned gate (G1), after which the pre-registration was frozen; model settings were never tuned on validation. Every later change is an amendment logged with its reason and date, made before the results it affects existed. Test labels are held behind a seal. Every configuration is scored on test without labels; a SHA-256 digest of the scores is logged with the single opening allowed per experiment, dataset and seed. A repeat is

allowed only after a crash, and is logged. The week-18 tests (HI- and LI-Medium) decided the pre-registered hypotheses.

D. Fresh replication (amendment A25)

Improvements made after the week-18 test could not be scored on that test without tuning on it. They were therefore developed on Medium validation only and selected by rules written before the candidates ran:

- *Protected flag*: among randomised-response and gated variants of the S1 flag, the one within the privacy margin on every target of both Medium sets with the largest share of the plain flag's gain. This selected the gated flag, with 1.01 of the gain.
- *Narrative*: among four local models (3–8 B parameters), the highest strict grounding; N4 built on it if at least as grounded. This selected N4 on `qwen3:8b`.
- *Benchmark adequacy*: if our pooled detector fell more than 5 F1 points below published LightGBM results on the benchmark's validation split, add a stronger detector as an extra track. It fell 14.5 points below on HI-Medium, so a transaction-level detector (Section V-B) was added (S0+, S2+, oracle+).

The owner then signed a frozen commit, and the two Large windows were opened once per experiment and seed: the main track, an artefact-free track and the stronger-detector track.

E. Statistics

Headline configurations ran with five seeds and the others with three; a seed fixes model training and negative sampling. Per-scheme outcomes are averaged over seeds before a cluster bootstrap over schemes (1,000 replicates). Gains are paired over the same schemes, with Holm correction [17] across the eleven sharing configurations compared with S0.

F. Leakage audit

A curious peer attacks everything it receives (seed 0 of each run). A1 re-identifies tokens from candidate lists of its own counterparties, under hashing and under the OPRF budget. A2 infers whether the queried account transacted with a third member; A3 infers top-decile outgoing amount, top-decile fan-in and pass-through behaviour. The A2/A3 attacker is a gradient-boosted classifier trained on the true attributes of half the queried accounts; the baseline is a prior-only classifier on what the peer knew before asking. The pre-registered margin requires the upper 95% bound of (attack − prior) true-positive rate at 1% false-positive rate to be at most 0.05 on every target, for S1 and for S3 at $\varepsilon \leq 2$.

V. Results

A. Detection

Fig. 1 and Table III summarise the sealed tests. Local monitoring links few schemes: 0.063–0.136 of them. Minimised evidence (S2) raises linked detection to 0.702 (HI-Medium) and 0.657 (LI-Medium) on the week-18 tests, and to 0.534 (HI-Large) and 0.450 (LI-Large) on the fresh tests; every gain has its 95% interval above zero after Holm correction, and

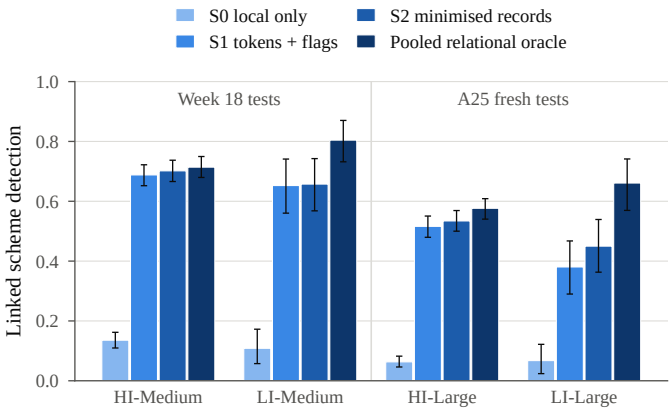


Fig. 1. Linked scheme detection on the four sealed tests by sharing level: local only (S0), tokens and alert flags (S1), minimised evidence records (S2), and a pooled relational oracle that sees every bank's data. Bars: mean over seeds; whiskers: 95% scheme bootstrap intervals. HI- and LI-Medium were decided on the week-18 tests; HI- and LI-Large are the A25 windows, never opened during development.

random alerts at the same budget link none. S2 recovers 0.98, 0.79, 0.92 and 0.64 of the relational oracle's gain, meeting the pre-registered 0.60 on all four tests, although the LI-Large interval reaches 0.52. Most of the value arrives with the alert flag alone (S1: 0.516 on HI-Large), as the gate result suggested: the gap is in linking alerts across banks, not in detecting them. Plain pooling is not the ceiling: one model on every bank's data, linked by the same rule, reaches 0.434 on HI-Large, below S2.

B. Robustness

Artefacts. The two strongest features in every model are AMLworld payment-format shares and self-transfers, which may be simulator quirks. Without them (Fig. 2), the gain is 0.438 on HI-Large and 0.210 on LI-Large, both with intervals above zero. On LI-Large the alert flag alone then carries little (S1 0.054) while the records keep 0.217: the minimised record, not the flag, carries the signal there.

A stronger detector. On AMLworld's published split, a transaction-level LightGBM on the transaction's own fields and both accounts' features (Table IV) outperforms both published gradient-boosting baselines on all four sets and the best published GNN on two. Used as the local detector (S0+), it leaves the sharing gain in place: 0.433 on HI-Large and 0.355 on LI-Large. Its pooled version, however, is much stronger on LI-Large (0.764), so S2+ recovers only 0.49 of the corresponding relational oracle's gain there—the one robustness rule that missed.

C. Privacy

Attribute leakage. On the high-illicit tests the alert flag and the LDP records stay within the margin. On the low-illicit tests they do not (Fig. 3): a peer that receives flags infers top-decile fan-in with an advantage of 0.092 on LI-Medium and 0.043 [0.031, 0.059] on LI-Large, because in that regime an alert is itself informative about an account's in-degree. S3 does not

TABLE III
PRE-REGISTERED CLAIMS ON THE SEALED TESTS

Pre-registered claim	HI-Medium	LI-Medium	HI-Large	LI-Large
H1a: S2 − S0 linked detection	0.57 [0.53, 0.61] ✓	0.55 [0.45, 0.63] ✓	0.47 [0.43, 0.51] ✓	0.38 [0.29, 0.48] ✓
H1b: share of oracle gain ($\geq$0.60)	0.98 [0.94, 1.03] ✓	0.79 [0.69, 0.88] ✓	0.92 [0.87, 0.97] ✓	0.64 [0.52, 0.77] ✓
H1c: plain flag, fan-in advantage	−0.002 [−0.008, 0.003] ✓	0.092 [0.079, 0.103] ✗	0.001 [−0.002, 0.003] ✓	0.043 [0.031, 0.059] ✗
gated flag (A25)	—	—	0.001 [−0.001, 0.003] ✓	0.004 [0.000, 0.007] ✓
H1d: PD gain share, least disclosure	0.98 at 0.07 ✓	0.99 at 0.01 ✓	0.95 at 0.03 ✓	0.81 at <0.01 ✓
H3a: attribution above baselines	✓	✗	✗	✗
H3b: narrative grounding ($\geq$0.95)	0.86 [0.81, 0.90] ✗	0.92 [0.84, 0.96] ✗	1.00 [0.98, 1.00] ✓	0.95 [0.87, 0.98] ✓
PT2: hashing ÷ OPRF re-id. ($\geq$10)	19.9 ✓	8.7 ✗	8.1 ✗	9.3 ✗

Sealed tests; 95% intervals from a scheme bootstrap. H1c: upper 95% bound of (attack − prior) TPR at 1% FPR at most 0.05 on every attack target (A2, A3); the value shown is the binding target, A3 fan-in, and the mark is the verdict over all targets and, for the plain flag, the S3 levels with $\varepsilon \leq 2$. H1d: PD configuration reaching $\geq$0.80 of S2's gain with the smallest share of S2's disclosed accounts. H3b on Medium: free-text narratives (A23); on Large: select-then-template (N4). Medium columns: week-18 tests; Large: A25 fresh tests.

TABLE IV
DETECTOR AGAINST PUBLISHED RESULTS

Model	HI-Small	LI-Small	HI-Medium	LI-Medium
Ours, account-window detector	48.20 ± 0.69	23.21 ± 2.80	42.67 ± 2.76	24.08 ± 2.01
Ours, transaction model	64.25 ± 1.24	**34.99 ± 1.51**	**67.01 ± 2.27**	34.34 ± 1.55
LightGBM + graph features [12]	62.86	20.83	59.48	20.85
XGBoost + graph features [12]	63.23	27.30	65.70	28.16
Multi-PNA + EU [12]	**68.16**	33.07	66.48	**36.07**

Minority-class F1 (%) on AMLworld's published split (all transactions, by day 60/20/20), threshold chosen on validation; ours: mean ± s.d. over 5 seeds; best per column in bold. Pooled data in every row (the published models see all banks).

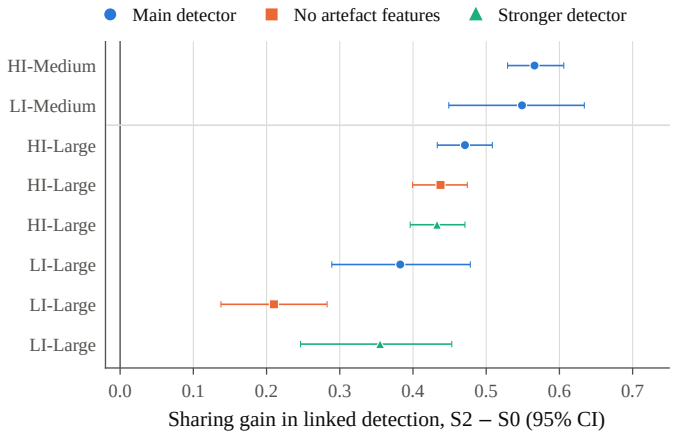


Fig. 2. Sharing gain (S2 − S0 linked detection) with 95% intervals. Main detector on all four tests; on the Large tests also without the suspected artefact features (payment-format shares, self-transfers) and with a stronger transaction-level detector that outperforms both published gradient-boosting baselines on the AMLworld benchmark (Table IV).

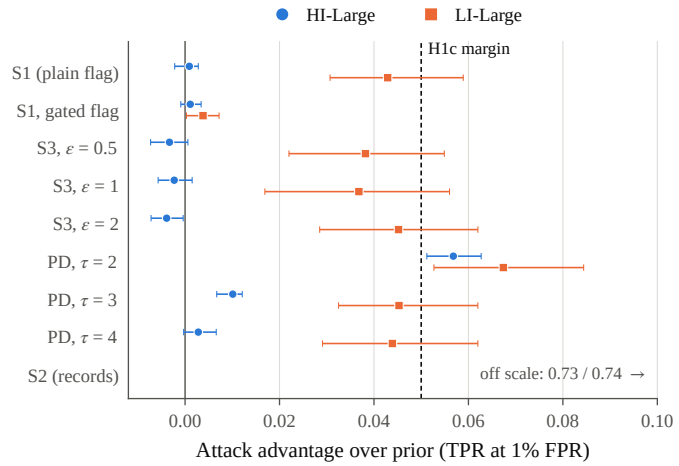


Fig. 3. Leakage audit on the Large tests: advantage of a strong attacker over a prior-only attacker in inferring whether a queried account has top-decile fan-in (the binding target), true-positive rate at 1% false-positive rate, with 95% intervals. The dashed line is the pre-registered margin (upper bound $\leq$ 0.05). On LI-Large the plain alert flag and every level that carries it exceed the margin; the gated flag chosen on validation does not.

help, since it does not perturb the flag. The gated flag, selected on validation, brings the advantage to 0.004 [0.000, 0.007] on LI-Large and keeps 1.00 (HI) and 0.93 (LI) of the plain flag's detection gain. Records (S2, S4) disclose attributes by design: the fan-in advantage is 0.73–0.74.

Progressive disclosure. PD at $\tau = 4$ keeps 0.95 (HI-Large) and 0.81 (LI-Large) of S2's gain while disclosing records on 3% and under 1% as many accounts. On the high-illicit sets this is uninformative, because the flag alone already reaches 0.95 of S2.

Tokens. Under hashing, a peer re-identifies 0.39 of other banks' record tokens on HI-Large; under the OPRF budget, 0.049. The ratio (8.1 on HI-Large, 9.3 on LI-Large, 8.7 on LI-Medium, 19.9 on HI-Medium) misses the pre-registered factor of 10 on three of four tests. Legitimate token use stayed within the frozen budget on both fresh tests. An adaptive budget (twice the previous epoch's use), replayed on recorded use, kept refusals far above its 1% target on busy days and lowered the ratio to 2.6–3.2 even under honest use; it was not adopted.

D. Case quality and analyst workload

For an investigator, a linked case should contain the scheme and nothing else. With sharing, the largest case holds 0.77 of

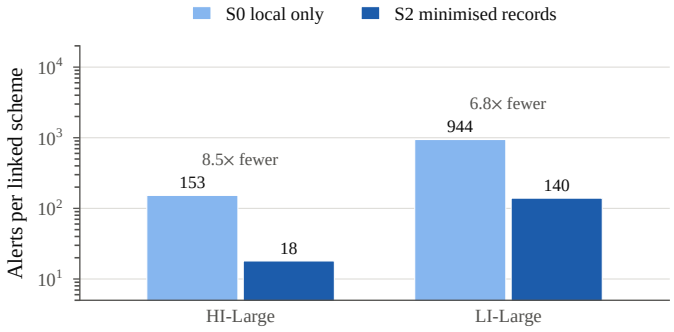


Fig. 4. Alerts an analyst reviews per laundering scheme linked, local-only monitoring (S0) versus minimised evidence sharing (S2), on the Large tests (logarithmic axis).

TABLE V
CASE QUALITY AND ANALYST WORKLOAD

Measure	HI-Large		LI-Large	
	S0	S2	S0	S2
Scheme accounts in the case	0.47	0.77	0.48	0.73
Cases per linked scheme	1.51	1.05	1.49	1.06
Alerts per linked scheme	153	18	944	140

Per detected scheme, seed-averaged: share of the scheme's accounts in its largest case (completeness), number of cases it is split into (fragmentation), and alerts raised per scheme linked (analyst workload). A25 fresh tests, seeds 0–4.

a scheme's accounts on HI-Large, against 0.47 locally, and a scheme is split into 1.05 cases instead of 1.51 (Table V). Because every bank keeps the same alert budget, the cost per linked scheme falls sharply: 18 alerts instead of 153 on HI-Large and 140 instead of 944 on LI-Large (Fig. 4).

E. Attribution and narratives

Typology attribution from fused evidence beat local attribution and both baselines on HI-Medium but not on LI-Medium or either fresh test; we treat it as exploratory. Free-text narratives from small local models were grounded in their structure—valid JSON, the right typology, existing fields—in 0.86 (HI-Medium) and 0.92 (LI-Medium) of first replies, short of the 0.95 target. Larger models (7–8 B) did not fix this: strict grounding, which also requires correct numbers, scope and claims, stayed at 0.06–0.22 on validation (Table VI). Select-then-template (N4) met the target on both fresh tests (1.00 and 0.95), and an automated judge rated its validation texts faithful in all 30 sampled cases, against 21–29 for free text.

VI. Limitations

AMLworld is synthetic; the results show what minimised sharing can achieve on the field's standard benchmark, not on real bank data. The low-illicit windows are small (78–91 cross-member test schemes), so their intervals are wide. The audit covers honest-but-curious peers and four attribute targets; collusion, poisoning and side channels are out of scope, and S1, S2 and PD responses are minimised and audited but not differentially private. The narrative judge is a single automated rater, neither blind nor independent of the template design. A federated, differentially private neural scorer, also

TABLE VI
CASE NARRATIVES: LOCAL MODELS AND SELECT-THEN-TEMPLATE

Design	Grounding	Strict	Faithful
HI-Medium validation, 200 cases (judge: 30)			
N0: `llama3.2:3b`, free text	0.96	0.13	21/30
N1: `qwen2.5:7b`, free text	0.97	0.10	26/30
N2: `llama3.1:8b`, free text	0.99	0.06	29/30
N3: `qwen3:8b`, free text	0.98	0.22	23/30
N4: select-then-template (N3)	0.99	0.83	30/30
N4 on the A25 fresh tests			
HI-Large	1.00	0.82	—
LI-Large	0.95	0.56	—

Grounding (H3b): first reply is valid, keeps the attributed typology and cites only evidence fields. Strict: also 1–4 fields, every number from the evidence, claim consistency and scope. Faithful: automated judge (one rater, not blind) on the 30-case sheet; delivered text after retry and template fallback.

pre-registered, met its target on LI-Medium but not on HI-Medium, where a strong local detector left little for it to add. Finally, every amendment after the pre-registration is logged with its reason. Two changed how token requests are metered so that runs would not stop on a token cap (A24a, A25a), both before the tests they affected were opened; one fixed a report page after an opening without changing any score (A24b).

VII. Conclusion

Sharing a little evidence about counterparties—a flag and a bucketed record, keyed by metered tokens—lets banks link several times more laundering schemes than local monitoring, close to what pooling all data would achieve, at a fraction of the alerts. The result held on four sealed tests, two of them never opened during development, without suspected artefacts and on a competitive base detector. Measuring leakage with attacks, rather than asserting it, found a real leak in the simplest disclosure and a fix that held on fresh data. We hope the evaluation discipline—pre-registration, sealed tests, logged openings and fresh replication—proves as useful to collaborative financial-crime research as the protocol itself.

Data and Code Availability

AMLworld is available under CDLA-Permissive-2.0. Code, the pre-registration, every amendment, the sealed-test access log and the result pages are in the project repository (MIT licence); every figure and table is regenerated by one command from the committed result pages.

References

[1] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018.

[2] E. Altman, J. Blanuša, L. von Niederäusern, B. Egressy, A. Anghel, and K. Atasu, "Realistic synthetic financial transactions for anti-money laundering models," in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023.

[3] A. Davidson, A. Faz-Hernández, N. Sullivan, and C. A. Wood, "Oblivious pseudorandom functions (OPRFs) using prime-order groups," RFC 9497, IETF, Dec. 2023.

[4] T. Suzumura, Y. Zhou, N. Baracaldo, G. Ye, K. Houck, R. Kawahara, A. Anwar, L. L. Stavarache, Y. Watanabe, P. Loyola, D. Klyashtorny, H. Ludwig, and K. Bhaskaran, "Towards federated graph learning for collaborative financial crimes detection," NeurIPS 2019 Workshop on Robust AI in Financial Services; arXiv:1909.12946, 2019.

[5] L. Zheng, J. Ni, C. Zobel, and J. R. Birge, "Networked markets, fragmented data: Adaptive graph learning for customer risk analytics and policy design," arXiv:2512.24487, 2025.

[6] S. Arora, A. Beams, P. Chatzigiannis, S. Meiser, K. Patel, S. Raghuraman, P. Rindal, H. Shah, Y. Wang, Y. Wu, H. Yang, and M. Zamani, "Privacy-preserving financial anomaly detection via federated learning & multi-party computation," arXiv:2310.04546, 2023.

[7] A. Abadi, B. Doyle, F. Gini, K. Guinamard, S. K. Murakonda, J. Liddell, P. Mellor, S. J. Murdoch, M. Naseri, H. Page, G. Theodorakopoulos, and S. Weller, "Starlit: Privacy-preserving federated learning to enhance financial fraud detection," Cryptology ePrint Archive, Paper 2024/090, 2024.

[8] M. S. I. Khan, A. Gupta, O. Seneviratne, and S. Patterson, "Fed-RD: Privacy-preserving federated learning for financial crime detection," arXiv:2408.01609, 2024.

[9] M. B. van Egmond, V. Dunning, S. van den Berg, T. Rooijakkers, A. Sangers, T. Poppe, and J. Veldsink, "Privacy-preserving anti-money laundering using secure multi-party computation," in *Financial Cryptography and Data Security (FC 2024)*, 2024, cryptology ePrint Archive, Paper 2024/065.

[10] M. Brand, H. Ivey-Law, and T. Churchill, "FinTracer: A privacy-preserving mechanism for tracing electronic money," Cryptology ePrint Archive, Paper 2023/649, 2023.

[11] BIS Innovation Hub, "Project Aurora: the power of data, technology and collaboration to combat money laundering across institutions and borders," Bank for International Settlements, Tech. Rep., May 2023.

[12] B. Egressy, L. von Niederäusern, J. Blanuša, E. Altman, R. Wattenhofer, and K. Atasu, "Provably powerful graph neural networks for directed multigraphs," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024.

[13] J. Blanuša, M. Cravero Baraja, A. Anghel, L. von Niederäusern, E. Altman, H. Pozidis, and K. Atasu, "Graph feature preprocessor: Real-time subgraph-based feature extraction for financial crime detection," in *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF '24)*, 2024, doi: 10.1145/3677052.3698674.

[14] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017.

[15] S. L. Warner, "Randomized response: A survey technique for eliminating evasive answer bias," *Journal of the American Statistical Association*, vol. 60, no. 309, pp. 63–69, 1965.

[16] C. Dwork, F. McSherry, K. Nissim, and A. Smith, "Calibrating noise to sensitivity in private data analysis," in *Theory of Cryptography Conference (TCC)*, 2006.

[17] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.